ELECTRONICS HARNESS BUILD PROCEDURE

OBJECTIVE:

To build the electronics jumper harnesses used to connect the longer underwater power harness to the individual drone solenoid/thrusters.

HARDWARE REQUIRED:

1. 2-Pin Female/Socket Waterproof connectors (4 total per assembly)
2. 8-Pin Male/Pin Connector kit (1 per assembly)
3. Pins for Male Connector kit (8 needed per kit)
4. Calipers
5. Bottle of Silicon Conformal Coating liquid
6. Heat shrink tubing for wires (sizes 3/16" – black, and 5/16" – red)
7. Solder Seal Wire connector (size 26-24) (8 total per assembly)
8. Green cable, 2.5 ft length (contains 4, paired wire sets)
9. Color zip tie flags
10. Voltmeter and testing cables

DURATION:

Variable

PART I – 8-PIN MALE CONNECTOR KIT END ASSEMBLY

NOTE
Procedure is written for assembly of 1 Solenoid Jumper Harness. Each jumper will provide power to 4 solenoid/thrusters. You will need to repeat this procedure up to 6 times if your craft has 24 thrusters.

1. Retrieve Green cable at a length of 2.5 ft

2. Using calipers, measure and mark ~40mm of length from one end

Figure 1

3. Strip the green sheath off, and then remove the underlying silver sheath (ref Figure 2)

Figure 2

4. Remove the remaining paper sheath and separate the 8 wires

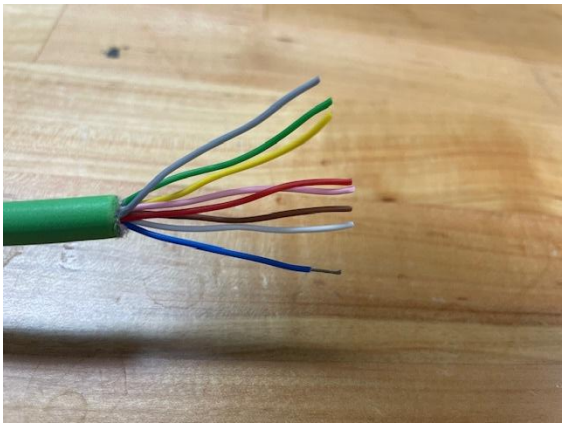

Figure 3 - (note blue wire has already been stripped)

5. Apply the silicon conformal coating liquid at the center base and around the perimeter where the wires meet the green sheath

6. Using the calipers, measure and mark ~4.8mm on each of the wires (ref Figure 4)

Figure 4

7. Strip that length of wire sheathing from each of the 8 pieces of wire

8. Watch this short video on crimping, note that there will be slight differences from the video

> CAUTION
> The pins tend to wobble on the crimping tool. Ensure that it is straight when you begin to pre-squeeze (prior to inserting the wire as per the video) the pin onto the crimping tool.

9. Hold crimping tool so that gauge labels on the tool head are facing you (Ref Figure 5)

10. Insert a pin (pin insertion will be a little tight) onto the 26/24 AWG (right) position of the crimping tool so that the side of pin where wire is inserted faces you. (Ref Figure 5)

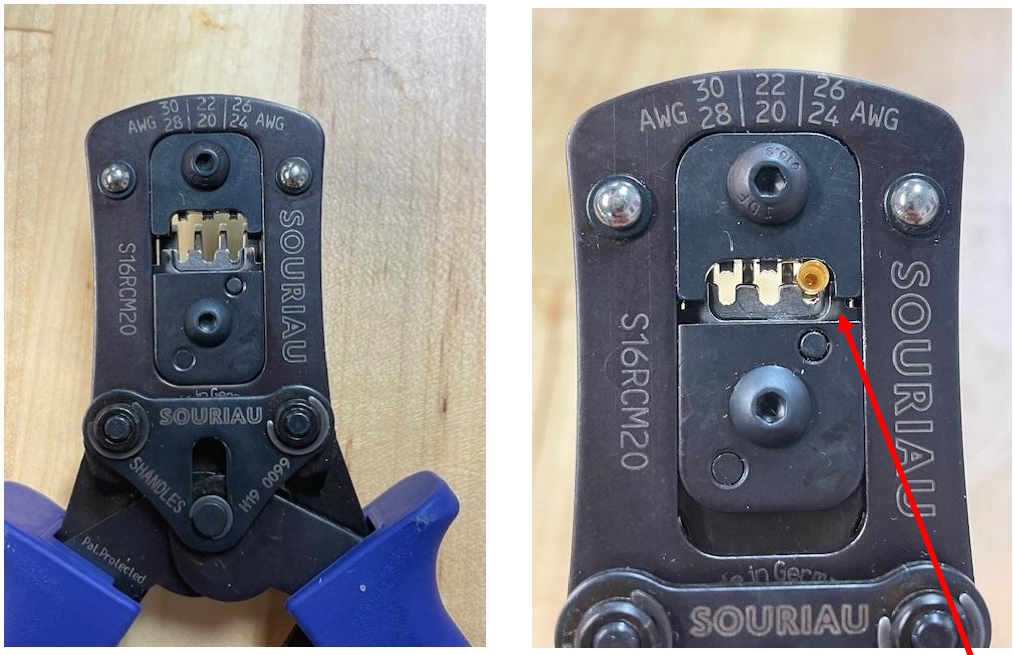

Figure 5

Pin already inserted in this picture, with side where wire inserts facing you.

11. Very slightly squeeze the crimper just enough to hold the pin in place (ref Figure 5). The crimping tool should <u>not</u> yet click.

12. Insert one of the 8 pieces of wire (exposed side) into the pin hole until it bottoms out.

13. While holding wire fully inserted into pin, strongly squeeze crimping tool, and release

<u>CAUTION</u>
Be cautious of wings on the pins during removal from the crimping tool.

14. Taking care not to damage the wings on the pin, remove the pin from the crimping tool

15. Once out of the tool, perform gentle tug on the wire/pin to ensure properly crimped onto the wire

16. Repeat steps 9 thru 15 for the remaining 7 wires

Figure 6

17. Retrieve the orange rubber grommet (Reference figure 6) - (you do not need the compression ring which is the hard black ring)

> <u>NOTE</u>
> For easier finger access, start step 17 with the Brown/Pin, into the center Slot H orange grommet spot.

18. Using the info below and Figure 7, **partially** insert each crimped pin into the orange rubber grommet – **insert from side with lettering** (you may have to use a very small flat head screwdriver and "break the orange seal" to help get started).

8-Pin Connector Wires
Red (Slot A)
Blue (Slot B)
Yellow (Slot C)
Green (Slot D)
Pink (Slot E)
Gray (Slot F)
White (Slot G)
Brown (Slot H)

For easier finger access start with the center Brown/H slot.

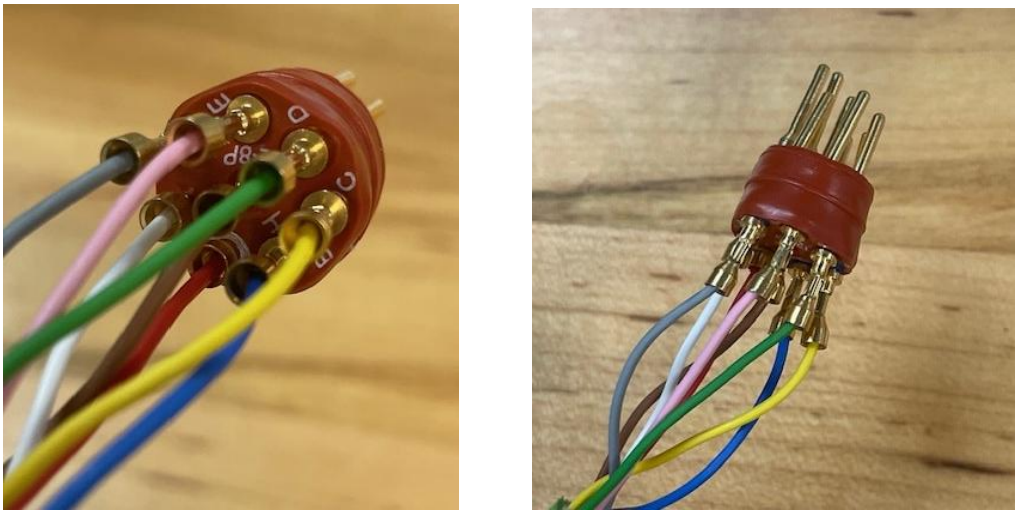

Figure 7

19. Complete insertion of all pins completely thru the grommet until around 0.75-1 inch of wire comes through (ref Figure 8)

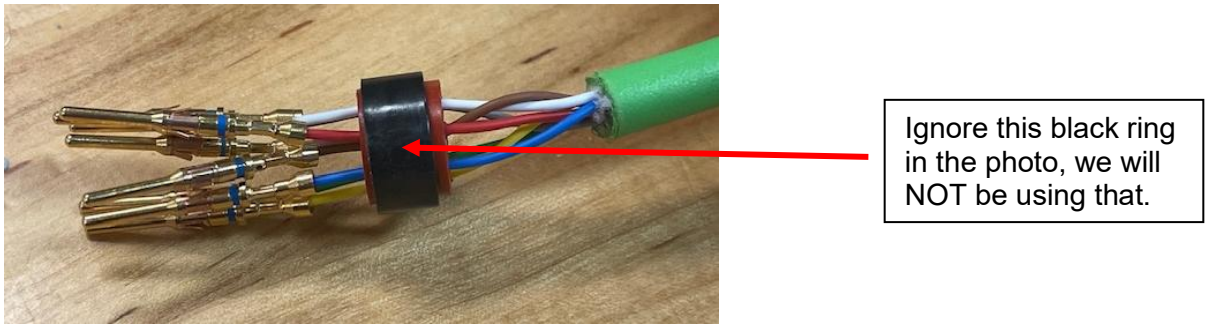

Figure 8 (black ring around orange grommet should not be present)

<u>CAUTION</u>
Ensure pins are inserted into connector fwd-shell starting from the end with the threads. Final product will have the pins recessed in the connector fwd-shell.

Once pins are fully inserted they lock into the fwd-shell and if inserted incorrectly the connector will be useless.

20. As needed, slide rubber grommet down toward the green sheath end to allow for more wire/pin access/flexibility

21. Using same configuration as listed in the table in step 18, and referencing the connector pin map in figure 9 below, <u>partially</u> insert all 8 pins into the connector fwd-shell

21.1. Recommend starting with the Center H/Brown wire slot for easier access

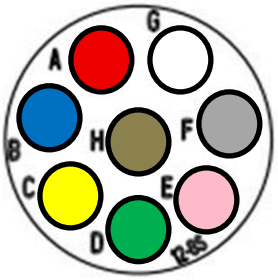

Figure 9 (Connector Pin Map)

22. Once all 8 have been started, using a small flat head screwdriver, push on the edge (NOT MIDDLE) of the back of the pins until they click/lock into place (ref Figure 10)

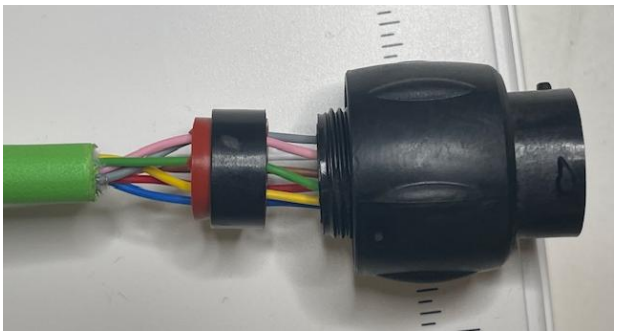

Figure 10 (black ring around orange grommet should NOT be present)

23. Perform gentle tug test on all wires, to ensure pins are locked

24. Slide rubber grommet up wire length until it is seated against fwd-shell (ref Figure 11)

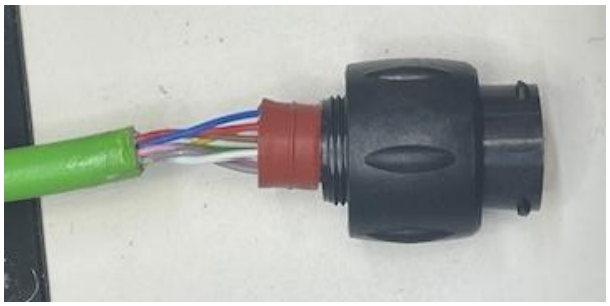

Figure 11 (rubber grommet up against fwd-shell)

25. Retrieve connector back-shell (ref Figure 12 for the various parts)

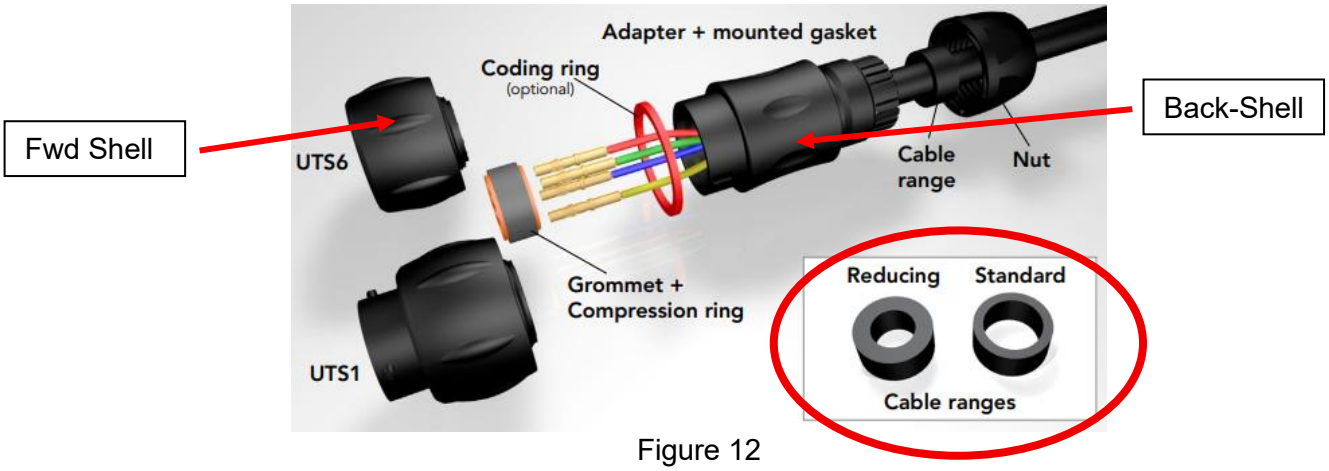

Figure 12

26. Carefully remove the “standard size” black rubber cable range/ring (ref Figure 12 – red circle)

27. Retrieve “reducing size” (thicker) black rubber cable range/ring from your kit

28. Insert thicker cable range/ring into back-shell until fully seated (ref Figure 13)

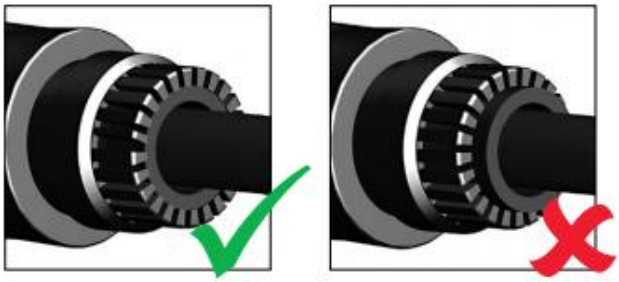

Figure 13 (Cable Range Proper Positioning)

29. Insert free end of green wire through connector back-shell

30. Slide back-shell up wire harness, and screw into connector fwd-shell

31. You may need to adjust the cable range/ring during and after connection of the fwd and back-shells.

32. Insert the connector nut through the free end of the green wire and screw into back-shell (ref Figure 14)

33. Ensure a tight connection at the fwd/back-shells and at the nut/back-shell (goal is for this to be waterproof)

Figure 14 (nut/back-shell connection tight)

34. Repeat all of PART I of this document to assemble additional Male Connector kits. Each connector assembly supports 4 thrusters. Depending on how many total thrusters you have, you may have to perform PART I, up to 6 times.

PART II – 2-PIN (FEMALE) END ASSEMBLY

1. Retrieve one of the 8-pin connector assemblies built in Part I of this procedure

2. Using wire stripers, remove ~6-in. of the external green sheath, and underlying paper cover, from the open end (ref Figure 15)

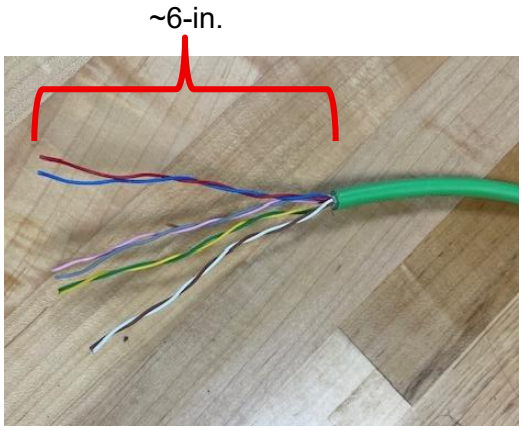

Figure 15

3. Splay the wire sets to allow access to the center for application of the silicon conformal coating liquid (ref. left photo in Fig 16)

4. Apply the silicon conformal coating liquid at the center base and around the perimeter where the wires meet the green sheath (both red circles in Fig 16)

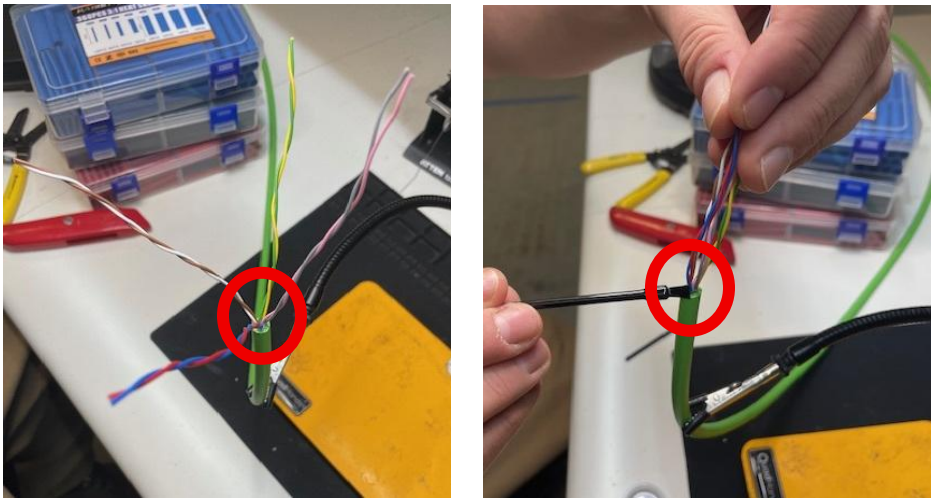

Figure 16

5. Repeat steps 1 thru 4 for all other 8-pin connector assemblies built in Part I of this procedure

6. Retrieve a piece of 3/16” heat shrink tubing (black), and cut to a length of 1/2 – 3/4 inches

7. Slide the heat shrink tubing over all 8 wires and down to the base at the green sheath

Figure 17

NOTE
The heat shrink tubing is marine grade and contains an additional sealant coating along the internal wall of the tubing. This coating liquifies and flows when heat is applied, helping to provide a more watertight seal.

After applying the heat gun, you should see the inner tubing sealant liquify and flow. Reference Figure 19

8. Using the heat gun, shrink the heat shrink tube.

9. Retrieve a piece of 5/16” heat shrink tubing (red), and cut to 1-1.25” length

10. Slide the heat shrink tubing over all 8 wires, and over the piece of 3/16” tubing just installed such that some overlap is present as shown in the left picture of Figure 18 below

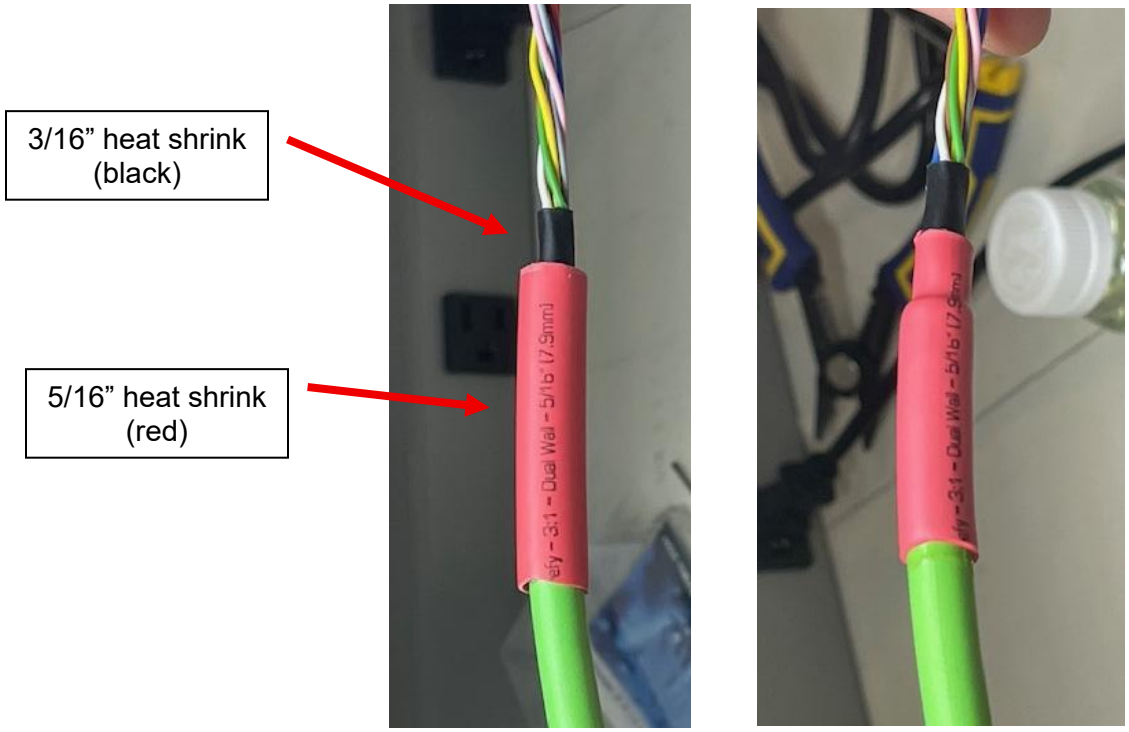

Figure 18 (before and after heat applied)

11. Using the heat gun, shrink the heat shrink tube.

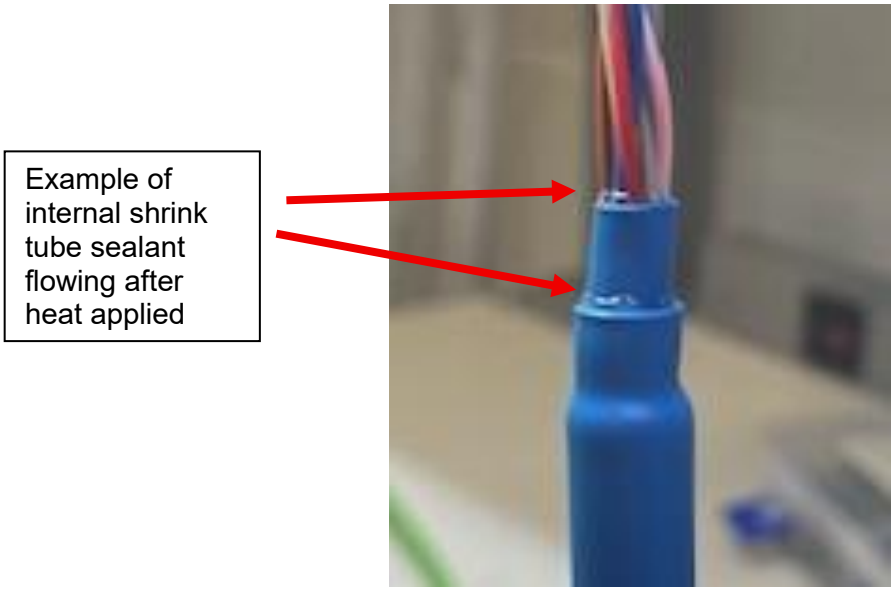

Figure 19

12. Set this assembly aside until step 20

13. Repeat steps 6 thru 12 to prep all other 8-pin connector assemblies built in Part I of this procedure

14. Retrieve a female 2-pin connector

15. Apply the silicon conformal coating at the base of the exposed wires (red circle in Figure 20)

> <u>NOTE</u>
> Some of the 2-pin connectors have been previously used, therefore you may have to strip off some of the outer-most black sheath to more represent the photo in Fig 20.

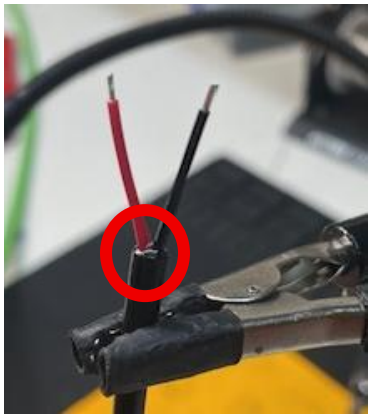

Figure 20

16. Retrieve a piece of 3/16” (black) heat shrink tubing, and cut to a length of ~1/2 inch

17. Slide the heat shrink tubing over the cables, overlapping with some of the black sheath (reference Figure 21)

18. Using the heat gun, shrink the heat shrink tube.

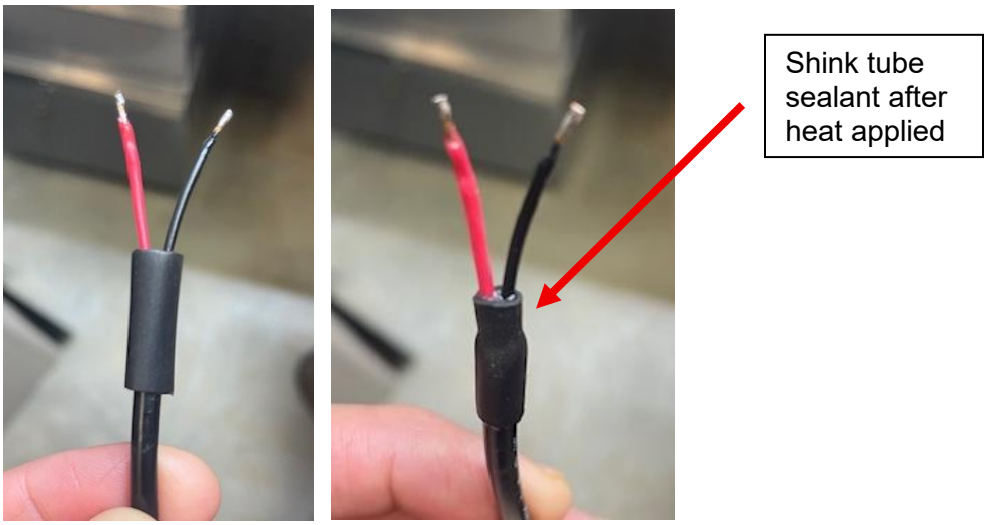

Figure 21 (before and after heat applied)

19. Repeat steps 14 thru 18 to prep all other female 2-pin connectors needed (up to 4 per jumper harness)

20. Retrieve an 8-wire assembly from step 12.

21. On each of the 8 wire ends strip ~4mm of sheath (use calipers and the 24-wire gauge slot on the wire stripers)

22. Retrieve a female 2-pin connector

CAUTION
Before connecting the 2-pin connectors to the 8-wire assembly, ensure that the cap is present on the 2-pin connector. Once the wires are mated, the cap cannot be installed.

Reference Figure 22 below

Figure 22 (Female 2-pin connector w/ cap present)

NOTE
The next steps will join the wire ends of the 2-pin and 8-pin cables. Instead of soldering, Solder Seal Wire Connector tubes will be used making the connections easier.

Steps 23 thru 27 are written for making the Connector 1, Black/Blue (Pin B) connection (per Table 1 below). These steps will be repeated for all wires/pins.

The 2-pin female connector wires and the 8-wire bundle will be joined per the below table.

Final 2-Pin Connector Reference Number	Zip Tie Flag Color	2-Pin Connector Wires	8-Pin Connector Wires
2-Pin Connector 1	Red	Red	Red (Pin A)
		Black	Blue (Pin B)
2-Pin Connector 2	Yellow	Red	Yellow (Pin C)
		Black	Green (Pin D)
2-Pin Connector 3	Orange	Red	Pink (Pin E)
		Black	Gray (Pin F)
2-Pin Connector 4	White	Red	White (Pin G)
		Black	Brown (Pin H)

Table 1

23. Retrieve a solder seal wire connector tube size 26-24 (reference Figure 23)

Figure 23 (Solder Seal Wire Connector tube size 26-24)

24. Insert the black wire end of a 2-pin connector into the solder seal wire connector tube so the exposed wire overlaps with the solder ring in the middle of the tube (ref Figure 24)

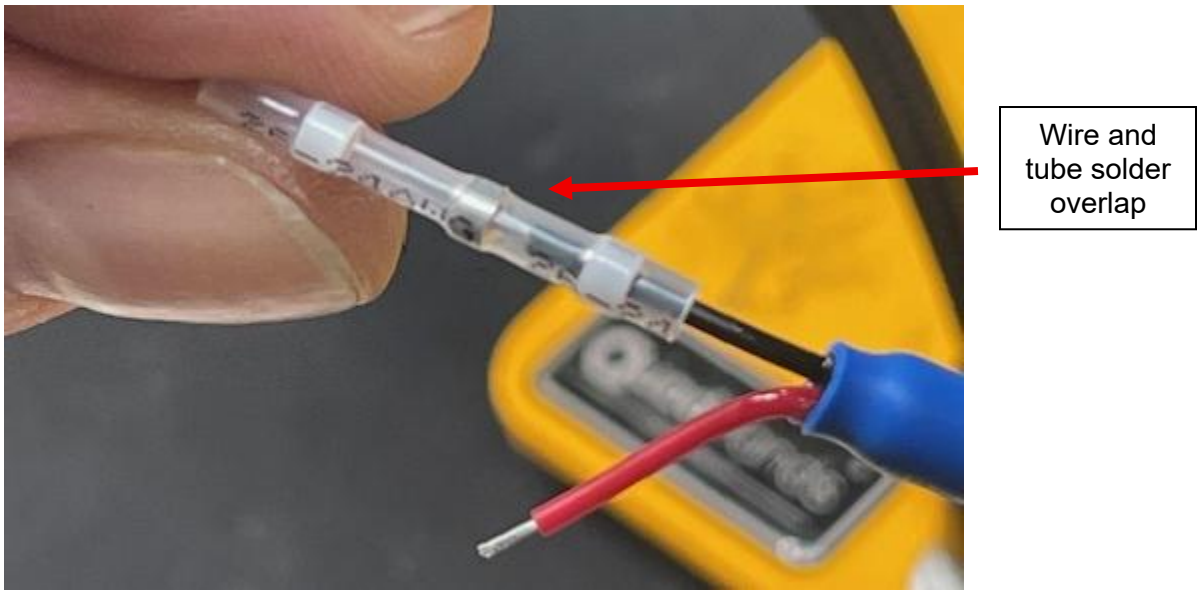

Figure 24 (2-pin connector wire within solder seal wire connector tube)

25. Insert the Blue exposed wire tip (8-wire harness side) to overlap with the solder seal wire connector tube solder ring, <u>as well as</u> with the exposed 2-pin black wire (ref Figure 25)

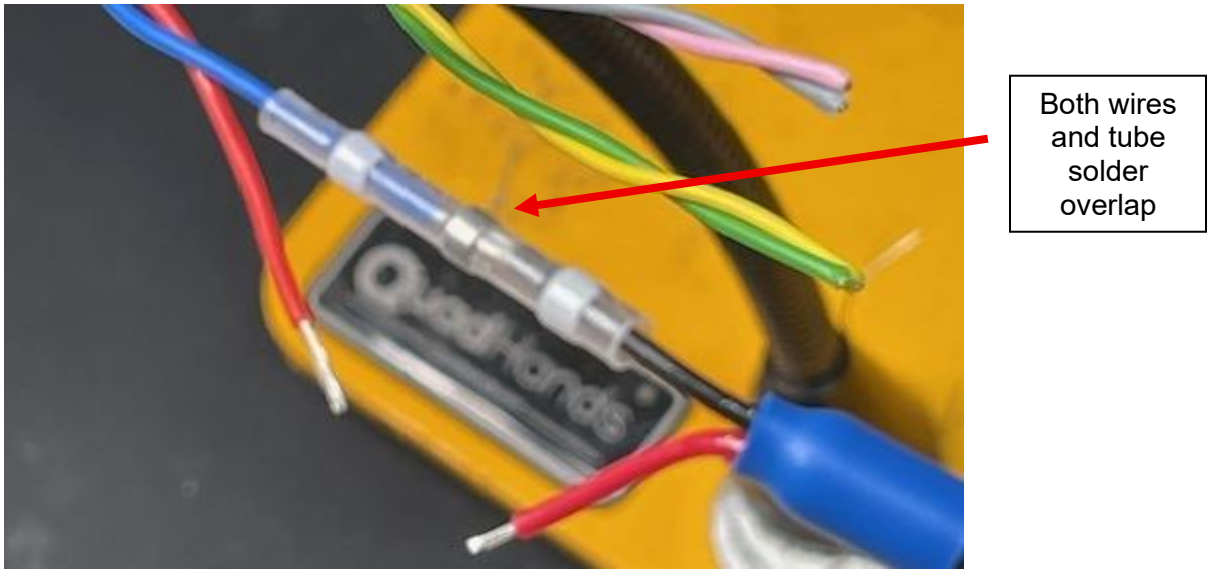

Figure 25 (Blue and Black wires overlapping with solder ring within tube)

NOTE
When heating the Solder Seal Wire Connector tube, the white rings on either end of the tubes will melt first. You must continue heating until the center solder ring melts and flows to make the electrical connection.

The flowing of the solder ring when melted will be visible

26. Using the heat gun, shrink the Solder Seal Wire Connector tube until the central solder ring on the tube flows.

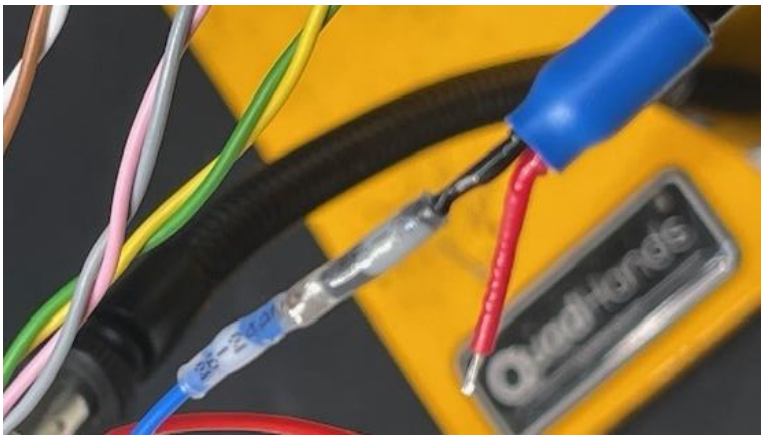

Figure 26 (Connection after heat applied)

27. Go to Part III of this procedure to use the Ohm Meter and confirm good connection before continuing.

28. Repeat steps 22 thru 27 to join the Red 2-pin connector end and the Red 8-pin connector wire end.

29. Install a red Zip Tie with a red flag on the 2-pin cable near the 2-pin connector end (ref Figure 27)

30. Connector #1 of a jumper harness (per Table 1) is now complete

31. Repeat steps 21 thru 30 to build Connectors #2, #3, and #4 for this jumper harness assembly joining wires and adding color zip tie flags per Table 1 (page 14)

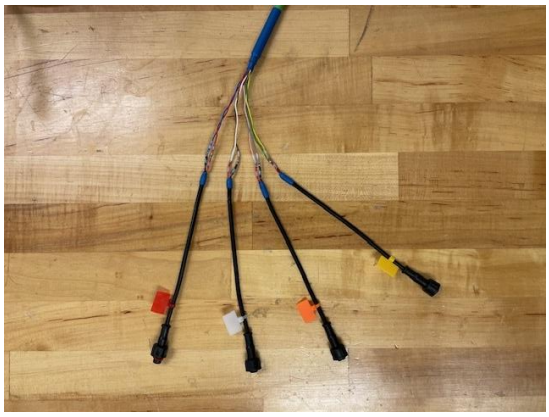

Figure 27 (final 2-pin connectors, w/ zip tie flags)

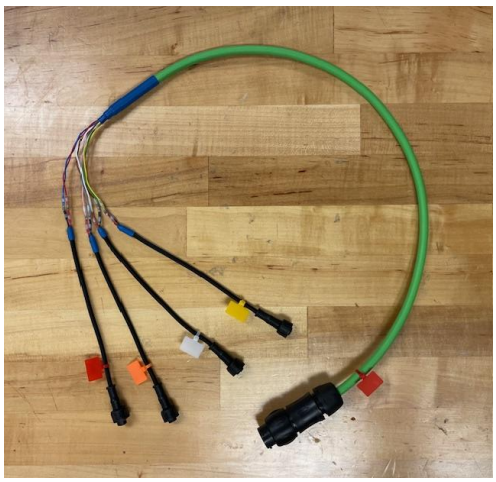

Figure 28 (Final built jumper harness assembly)

32. Repeat steps 20 thru 31 to add 2-pin connectors to any other 8-pin connector assemblies. Each connector assembly supports 4 thrusters. Depending on how many total thrusters you have, you may have to repeat up to 6 times.

33. Once complete with all jumper harnesses, add color Zip Ties flags at the 8-pin connector end per the Table 2 below (ref Figure 28)

Jumper Harness 8-Pin Connector Zip Tie Color References	Zip Tie Flag Color
Connector 1	Red
Connector 2	Blue
Connector 3	Green
Connector 4	Yellow
Connector 5	Orange
Connector 6	Black

Table 2

PART III – TESTING YOUR CONNECTIONS

1. Retrieve a Voltmeter (reference Figure 29)

2. Set the dial to the Ohm (straight up) position (reference Figure 29)

3. Push the yellow button twice until you see the sound icon at the top of the screen (reference Figure 29) – if you hear a beep during testing, you have good continuity

Figure 29

NOTE
If you are checking for continuity on a male vs. female pin, you may need to use two different connector ends (ref Figure 30).

Figure 30 – left picture for male pins, right picture for female pins

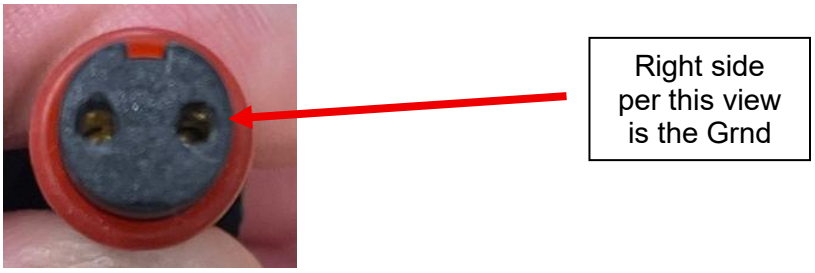

Figure 31 (female end of 2-pin connector – Red/Pwr left side, Black/Grnd right side)

4. Place one needle tip from the voltmeter cable to one of the 8-pin connector pins (ref Table 1)

5. Place other needle tip (may need to use small wire vs. needle) into the corresponding female end of the 2-pin connector. You should hear a continuous beep from the voltmeter if you have good connectivity.

Final 2-Pin Connector Reference Number	Zip Tie Flag Color	2-Pin Connector Wires	8-Pin Connector Wires
2-Pin Connector 1	Red	Red/Pwr/Left	Red (Pin A)
		Black/Grnd/Right	Blue (Pin B)
2-Pin Connector 2	Yellow	Red/Pwr/Left	Yellow (Pin C)
		Black/Grnd/Right	Green (Pin D)
2-Pin Connector 3	Orange	Red/Pwr/Left	Pink (Pin E)
		Black/Grnd/Right	Gray (Pin F)
2-Pin Connector 4	White	Red/Pwr/Left	White (Pin G)
		Black/Grnd/Right	Brown (Pin H)

Table 1